

Haokai Ding

Shenzhen, China

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Summary

Undergraduate researcher in robotics, focusing on underactuated grasping, idle-stroke and Peaucellier-based linkages, and contact-rich aerial manipulation. Building hardware prototypes and analytical models that bridge mechanism design with field-deployable manipulation. Incoming MSc in Robotics at MBZUAI.

Education

Mohamed bin Zayed University of Artificial Intelligence (MBZUAI)

M.SC. IN ROBOTICS (INCOMING)

Abu Dhabi, UAE

Fall 2026

Tsinghua University | Shenzhen X-Institute

TSIEN EXCELLENCE IN ENGINEERING PROGRAM (TEEP) – X SCHOLAR

Shenzhen, China

Sep. 2023 – Present

- Joint training program in underactuated robotics and mechanism design.

Shenzhen Technology University – Future Technology School

B.ENG. IN ELECTRONIC SCIENCE AND TECHNOLOGY

Shenzhen, China

Sep. 2022 – Jul. 2026

- GPA: 84.6/100. Research focus: LIBS spectroscopy and robotics systems.

Research Experience

State Key Lab of Mechanical Systems & Vibration, Shanghai Jiao Tong University

VISITING STUDENT – UAV PERCHING AND AERIAL MANIPULATION

Shanghai, China

Feb. 2026 – Present

- Advisor: Prof. Wei Dong.
- Working on UAV perching and contact-rich aerial manipulation.

Shenzhen X-Institute, Tsinghua University

UNDERGRADUATE RESEARCHER – UNDERACTUATED ROBOTIC GRIPPERS

Shenzhen, China

2023 – Present

- Designed and tested underactuated grippers based on semi-Peaucellier linkages.
- Developed idle-stroke and delay-triggering mechanisms; validated prototypes via simulation and physical experiments.

Shenzhen Technology University

UNDERGRADUATE RESEARCHER – LIBS SPECTROSCOPY FOR HEAVY-METAL ANALYSIS

Shenzhen, China

2022 – 2023

- Built and optimized a LIBS system for rapid heavy-metal adsorption analysis.
- Improved the calibration pipeline and acquisition accuracy.

Publications

† Co-first author * Corresponding author Listed in reverse chronological order by IEEE Xplore date added.

SELECTED CONFERENCE PAPERS (ROBOTICS)

[P1] An Idle-Stroke Underactuated Gripper with Independent Double-Parallelogram Linkages for Pinching and Adaptive Grasping

Y. CHEN, H. LUAN, **H. DING**, W. ZHANG* | *IEEE RAAI 2025, Singapore* | Mar. 2026

[P2] BioCrest – A Flexible Adaptive Robotic Hand Based on Idle-Stroke Mechanism

H. DING†, X. ZHONG, W. HUANG, T. YANG*, W. ZHANG* | *IEEE ROBIO 2025, Bangkok, Thailand* | Feb. 2026

[P3] GLINT: An Idle-Stroke Grasp-and-Lift Hand for In-Hand Manipulation

H. DING, H. LUAN, T. YANG*, W. ZHANG* | *IEEE ICCMA 2025, Macau, China* | Feb. 2026

[P4] Peaucellier Gripper: A Novel Underactuated Gripper for Linear Pinching and Self-Adaptive Grasp

H. DING†, J. FAN, Z. ZHU, Y. ZHAO, K. CHEN*, W. ZHANG* | *IEEE ICARM 2025, Hangzhou, China* | Dec. 2025

[P5] An Underactuated Gripper with Grasshopper-Inspired Linkages and Delay Triggering for Pinching and Adaptive Grasping

H. DING†, Y. CHEN†, D. JIA, W. ZHANG* | *IEEE ICIA 2025, Wuhan, China* | Dec. 2025

[P6] **A Novel Gripper with Semi-Peaucellier Linkage and Idle-Stroke Mechanism for Linear Pinching and Self-Adaptive Grasping**
H. DING[†], W. ZHANG^{*} | *IEEE/RSJ IROS 2025, Hangzhou, China* | Nov. 2025

[P7] **Semi-Peaucellier Linkage and Differential Mechanism for Linear Pinching and Self-Adaptive Grasping**
H. DING[†], Z. CHEN, T. YANG^{*}, W. ZHANG^{*} | *IEEE CASE 2025, Los Angeles, USA* | Sep. 2025

OTHER PUBLICATIONS

[P8] **Recent Progress on the Research of 3D Printing in Aqueous Zinc-Ion Batteries**
Y. LIU[†], H. DING[†], H. CHEN, H. GAO, J. YU, F. MO, N. WANG^{*} | *Polymers (MDPI) 17(15):2136* | Aug. 2025

[P9] **Rapid Analysis of Heavy-Metal Element Adsorption by SCG Based on LIBS Technology**
H. XIE, L. YOU, X. FANG, L. LI, H. DING, Z. ZHOU, G. ZHANG, D. ZHANG | *E3S Web of Conferences, 615 (2025) 02010* | Feb. 2025

Honors & Awards

2025	Undergraduate Champion , ASME Student Mechanism & Robot Design Competition	USA
2025	National Scholarship , Ministry of Education of China	China
2024	President’s Award , Shenzhen Technology University	Shenzhen, China
2024	Research and Innovation Award (First Prize) , Shenzhen Technology University	Shenzhen, China
2023	Gold Prize , 9th China “Internet+” Innovation Competition (Guangdong Division)	Guangdong, China
2023	Second Prize , 17th “Challenge Cup” Extracurricular Science Competition (Guangdong Division)	Guangdong, China
2023	Research and Innovation Award (First Prize) , Shenzhen Technology University	Shenzhen, China

Professional Service

2025	Conference Reviewer , IEEE/RSJ International Conference on Intelligent Robots and Systems (IROS)	
2025	Session Chair , IEEE/RSJ International Conference on Intelligent Robots and Systems (IROS)	Hangzhou, China
2025	Conference Reviewer , IEEE International Conference on Automation Science and Engineering (CASE)	